Neel Patel

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.19
The H B Kapadia New High SCH	Class XII (CBSE)	2024	88.40%
Shanti Asiatic School	Class X (CBSE)	2022	95.80%

EXPERIENCE

Product Intern | Vetifi Technologies Pvt. Ltd.

- **Auditing** existing user flows leading to an average **improvement** of up to **90** seconds by identifying crucial friction points in the **user onboarding** and **login verification** processes for veterinary practitioners and medical students.
- **Partnering** with the AI team on **LLM and RAG pipeline evaluation**, reviewing model outputs across **150+ test queries** spanning common veterinary diagnostic cases where predictions directly impact the treatment decisions.
- **Bridging workflow** across AI, tech, and design teams by authoring product specs on **Confluence** informed by direct feedback from **15+ veterinary doctors** and translating AI research outputs into actionable feature specifications.

PROJECTS

Physics Informed Neural Network for Phase Field Equations | Python, PyTorch, Matplotlib, SciPy | GitHub

- Implemented a **meshfree PINN** that solves Allen-Cahn PDE governing phase evolution and microstructure dynamics.
- **Trained** the model using PDE residual, initial condition, and boundary condition losses, requiring **no simulation data** and thus enabling deployment across practical phase field problems.
- Achieved just **3.5%** relative **L2** error against a classical Finite Difference reference solver.

IKEA Market Strategy | TAM/SAM/SOM Modeling, Comp. Benchmarking, Stress-Testing | View Slide/Excel

- Built a **TAM/SAM/SOM model** for **3** Tier-2 Indian cities from district income and population data, sizing a **178-317 Cr** serviceable market and a **14-25 Cr/year** revenue opportunity per city.
- **Stress-tested** unit economics against **12-18 Cr** in estimated fixed costs, finding only **1 of 3** cities cleared breakeven. Driving a risk adjusted entry sequence over the figured highest revenue option among the same.
- Delivered an **8-slide strategy deck** backed by a linked **Excel model** with weighted scoring, financial calculations, and fully source labeled assumptions from **3+** trusted sources (Knight Frank, company filings, government data).

Solving The Doppelgänger Problem in Microstructures | Python, pandas, Matplotlib | GitHub

- **Investigated** microstructures generated by fundamentally different physics, Cahn-Hilliard vs Allen-Cahn dynamics.
- **Quantified morphological separability** across **40** microstructure samples using centroid-based separation ratio of **1.38**, confirming that PCA features expose true structural differences.
- Generated a **32.5%** improvement in **classification accuracy** by PCA based extraction over standard baselines.

SCHOLASTIC ACHIEVEMENTS

National Standard Examination in Physics (NSEP), 2023 | View certificate

- Secured a **top 10 percentile** rank in the exam which emphasized multi-step advanced analytical physics problems.

Selected for Undergraduate Project | IIT Hyderabad

- Got selected by Dr. Saswata Bhattacharya (**HoD**, Materials Science & Metallurgical Engineering) to pursue a long-term research project on **Machine Learning applications** aligned towards the field of computational materials science.

Scored 100/100 | Maths 10th CBSE (2022); Received offer to BITS Pilani-B.E. Electrical Engineering (2024)

POSITIONS OF RESPONSIBILITY

Head of Multimedia and Design | E-Cell

- Managing **sponsor-facing** communications and **on-site** event branding materials by an expanded team of **32** individuals, ensuring consistent professional standards to attract and retain high value partnerships.

Head of Multimedia | E-Cell

- Led a team of **21** members in content strategy and branding, establishing E-Cell as a **credible** entrepreneurial platform and successfully attracting Upasana Kamineni (Vice Chairperson, **Apollo Hospitals**) as a speaker in a flagship event.
- Built E-Cell's digital presence across LinkedIn and Instagram, driving **200k+** organic impressions in just **2** months, generating inbound corporate interest and sponsorship visibility.

Class Representative | IIT Hyderabad

- Elected to represent **34** students, acting as the primary point of contact between academic administration and peers.

SKILLS/RELEVANT COURSES

Languages and Libraries: C++, Python, SQL, PyTorch, NumPy, Pandas, Matplotlib, SciPy

Courses: Linear Algebra, Numerical Methods, Differential equations, Computational Methods in Materials, Machine Learning and Data Analytics, Fundamentals of Scientific Computing, Calculus I, Calculus II

Concepts: Product Prioritisation (RICE, MoSCoW), A/B Testing, Physics-Informed ML, Neural Networks, DSA

Tools: Git, Jupyter Notebook, Excel, PowerBI, Confluence

EXTRACURRICULAR ACTIVITIES

- Represented **IITH** at the **Inter IIT Cultural Meet 8.0**, securing **4th position** among all IITs in photography.
- NSO **Badminton** player and **chess player**, reflecting a strong interest in strategic and game theory thinking; studying the book ***A Practical Guide to Quantitative Finance Interviews*** for mathematic problem solving foundations.
- **Head**, InFocus (IITH's photography club) | showcased composition skills over **8** events with **40k+** footfall.